

ROHITH PADYATHARA KUNJAPPAN

Generative AI & Machine Learning Engineer | Agentic AI Specialist

padiyathara3@gmail.com | 857-578-8130 | [LinkedIn](#)

PROFESSIONAL SUMMARY

- Results-driven **Generative AI and Machine Learning Engineer** with **5+ years of experience**, transitioning from a **real-time data engineering foundation** across banking and healthcare pipelines into designing, building, and contributing to **LLM-powered products, multi-agent AI systems, RAG pipelines, and production ML platforms**.
- Practical experience with **LLMs (GPT-4, Claude)**, **Retrieval-Augmented Generation (RAG)**, and **Agentic AI orchestration** using **LangChain and LangGraph**, with working knowledge of **Model Context Protocol (MCP)** for tool and context integration in agentic workflows.
- Experience building AI-enabled backend and full-stack applications with **Python, FastAPI, Flask, React.js, and Node.js** connecting backend microservices and AI inference layers with enterprise frontend systems.
- Skilled in building production-grade **fraud detection, anomaly detection, and predictive analytics models** using **XGBoost, Random Forest, and Scikit-learn**, consistently driving measurable precision, recall, and operational efficiency gains.
- Delivered **semantic search and vector retrieval systems** using **pgvector** in production, with additional hands-on experience evaluating **FAISS and Pinecone**, supporting AI-assisted investigation, risk-scoring, and clinical knowledge-retrieval platforms.
- Experience deploying containerized AI workloads using **Docker and event-driven Kafka pipelines**, with infrastructure automation via **CI/CD (GitHub Actions, AWS CodePipeline)** and **CloudFormation** on AWS.
- Familiar with **MLOps/LLMOps practices, LLM evaluation (Ragas, TruLens), AI guardrails, model monitoring, drift detection, and Explainable AI (XAI/SHAP)**, supporting governance and transparency in regulated, production-grade AI deployments.
- Comfortable in Agile/Scrum delivery and cross-functional collaboration with product, data science, and compliance teams; experienced with **Git, GitHub, JIRA, and performance benchmarking** in 24x7 production environments.

TECHNICAL SKILLS

Generative AI & LLMs	GPT-4, Claude, OpenAI API, Anthropic Claude API, Prompt Engineering, Retrieval-Augmented Generation (RAG), Embeddings, Function/Tool Calling, Structured Outputs
Agentic AI & Orchestration	LangChain, LangGraph, LangSmith, Model Context Protocol (MCP), Multi-Agent Systems, Tool-Augmented Reasoning, Human-in-the-Loop
AI/ML & Modeling	Scikit-learn, XGBoost, Random Forest, Time Series Modeling, Feature Engineering
Vector Databases & Semantic Search	pgvector, FAISS, Pinecone, Semantic Search Pipelines, Embedding Indexing, Hybrid Retrieval
Data Engineering & Processing	Apache Spark, PySpark, Kafka, Kafka Connect/Streams, Change Data Capture (CDC), Oracle GoldenGate, Airflow, Hive, Sqoop, ETL/ELT Pipelines, HL7/FHIR, Batch & Streaming Data Systems
Backend & API Development	FastAPI, Flask, REST APIs, Microservices Architecture, Async Processing, JavaScript (ES6+), React.js, Node.js, Express.js
MLOps & LLMOps	Docker, MLflow, CI/CD Pipelines, Model Monitoring, Data Drift Detection, LLM Guardrails (Guardrails AI), Eval Pipelines (Ragas, TruLens), Responsible AI
Cloud Platforms (AWS)	AWS Bedrock, Bedrock Agents, SageMaker, Lambda, ECS, EC2, S3, RDS, IAM, VPC, KMS, CodePipeline, CloudFormation, CloudWatch, Kinesis
Databases & Storage	PostgreSQL, Oracle, Hive, SQL / PL-SQL
Security & Explainability	Explainable AI (XAI), SHAP, IAM, VPC, KMS, PII Compliance, Data Governance
Data Visualization & BI	Power BI, Tableau
DevOps & Automation	AWS CodePipeline, CloudFormation, Jenkins, Git, GitHub, GitHub Actions, Agile/Scrum

PROFESSIONAL EXPERIENCE

Generative AI & Machine Learning Engineer — Banking Fraud & Risk Analytics Platform

April 2024 – Present

Cognizant

- Built and deployed a **real-time AI-powered fraud detection and credit risk scoring pipeline** using **Python, Scikit-learn, XGBoost, and FastAPI** to flag high-risk banking transactions, achieving **93% precision** through optimized XGBoost and Random Forest classifiers.
- Extracted high-signal fraud features spending velocity, geolocation variance, device fingerprinting, failed login frequency, and abnormal transaction amounts and tuned precision, recall, and F1 metrics, **reducing false-positive rates by approximately 28%** through threshold calibration.
- Developed **REST APIs using FastAPI** for real-time fraud scoring and built an **LLM-powered fraud investigation assistant using LangChain and LangGraph** to automate alert triage and support analyst reasoning, **cutting manual review time by roughly 32%**.
- Connected **OpenAI and Anthropic Claude LLM APIs to RAG pipelines with pgvector** to surface historical fraud patterns through semantic search, powering AI-assisted case investigations on top of a **streaming-ready Kafka infrastructure** for high-volume alert ingestion.
- More recently, introduced **Ragas and Guardrails AI** for LLM output evaluation and validation, and began adopting **Model Context Protocol (MCP)** to standardize tool and context access across agent workflows as the team's agentic tooling matured reducing hallucination rates and supporting auditable model behavior.
- Built model monitoring and explainability dashboards using **Python, SQL, and SHAP values** to track fraud trends, anomaly/data-drift rates, and model transparency for compliance reviews; containerized ML and LLM inference services using **Docker** with **MLflow** experiment tracking for reliable production deployments.
- Collaborated with frontend and DevOps engineers to integrate fraud-scoring APIs into **React.js-based investigation dashboards** (reducing analyst turnaround time by **over 24%**) and to deploy inference services on **AWS (SageMaker, Lambda, Bedrock Agents)** through **CodePipeline/CloudFormation** CI/CD automation.

Environment: Python, FastAPI, Scikit-learn, XGBoost, Random Forest, OpenAI API, Anthropic Claude API, LangChain, LangGraph, MCP, RAG, pgvector, PostgreSQL, SQL, Kafka, Docker, REST APIs, React.js, Node.js, AWS (Bedrock, Bedrock Agents, SageMaker, Lambda, CodePipeline, CloudFormation, CloudWatch), Pandas, NumPy, SHAP, Git, GitHub, Microservices Architecture

Data Engineer — AI/ML Integration, Real-Time Healthcare Analytics

January 2021 – December 2021

Apollo

- Built **real-time data pipelines** ingesting patient vitals, admissions, lab results, and bedside monitoring-device data using **Apache Kafka and Spark Structured Streaming**, processing **HL7/FHIR-formatted clinical messages** with sub-minute latency to feed downstream AI/ML scoring.
- Contributed to feature engineering for **real-time patient-risk indicators** (vitals anomaly scores, readmission-risk indicators, length-of-stay predictors) alongside the data science team, and supported training and validation of early **patient-risk-stratification models** using Scikit-learn and XGBoost, gaining hands-on exposure to applied AI/ML work.
- Supported integration of a trained **anomaly-detection ML model** into the real-time streaming pipeline, helping replace static threshold-based vitals alerts with model-driven scoring, which **contributed to a reduction of approximately 20% in false alert volume** for nursing staff while preserving early-warning sensitivity.
- Assisted with the model's lightweight **MLOps lifecycle**, supporting versioning and experiment tracking in **MLflow**, monitoring precision/recall and prediction drift in production, and helping coordinate scheduled retraining.
- Contributed to a **clinician feedback loop** that labeled false positives/negatives on model-generated alerts, feeding validated outcomes back into the training set to improve model precision over successive iterations.
- **Designed and maintained streaming and batch ETL pipelines** using Python, PySpark, and Airflow to transform raw HIS/EHR data into curated, model-ready and analytics-ready datasets in PostgreSQL and a cloud data warehouse.
- Built and exposed **REST APIs using FastAPI and Flask** to serve real-time model scores and operational metrics to **Power BI and Tableau** clinical dashboards for decision-makers.
- Supported deployment of streaming and ML inference services on **AWS (Lambda, S3, RDS, ECS, Kinesis)** with HIPAA-aligned security controls; used **AWS IAM, VPC, and KMS** for access control and data governance.

Environment: Apache Kafka, Spark Structured Streaming, PySpark, Airflow, HL7/FHIR, Scikit-learn, XGBoost, MLflow, Model Monitoring, Drift Detection, FastAPI, Flask, PostgreSQL, AWS (Lambda, S3, RDS, ECS, Kinesis, IAM, VPC, KMS), Power BI, Tableau, Pandas, NumPy, Git, Agile/Scrum

Data Engineer — Real-Time Core Banking Data Pipelines

May 2019 – December 2020

Genpact

- Contributed to building and maintaining **real-time and batch data pipelines** for core-banking transaction, settlement, and reconciliation data using **Apache Kafka, Spark Streaming, and Sqoop**, ingesting data from Oracle and mainframe-fed systems into a central data warehouse.
- Assisted in designing and implementing **Change Data Capture (CDC) pipelines** using **Oracle GoldenGate and Kafka Connect** to replicate transaction and account data in near real time, helping reduce data latency for downstream reporting from hours to minute.
- Contributed to **ETL/ELT workflows** using **Informatica PowerCenter and Spark (PySpark)** to transform and load high-volume transactions, customer, and loan data into Hive and Oracle data marts, supporting regulatory and operational reporting.
- Supported **real-time transaction reconciliation and exception-detection pipelines** using **Kafka Streams and Spark Structured Streaming**, helping flag settlement mismatches and duplicate postings within seconds of ingestion.
- Helped optimize **SQL and PL/SQL batch jobs** and Oracle/Hive partitioning strategies, contributing to **cutting nightly batch-processing windows by approximately 30%** and improving data availability for downstream banking applications.
- Assisted in implementing **data quality and validation checks** (schema validation, null/duplicate detection, reconciliation counts) across ingestion pipelines, contributing to **roughly a 25% reduction in data-defect escapes** to production reporting.
- Collaborated with backend and BI teams to expose curated datasets via **REST APIs and Power BI/Tableau** extracts for customer, transaction, and loan-servicing reporting.
- Engaged actively in **Agile/Scrum** ceremonies sprint planning, backlog refinement, code reviews, and testing coordination to support consistent, high-quality iterative delivery.

Environment: Apache Kafka, Kafka Connect, Kafka Streams, Spark (Core, Streaming, PySpark), Oracle GoldenGate, Informatica PowerCenter, Sqoop, Hive, Oracle, PL/SQL, SQL, Airflow, REST APIs, Power BI, Tableau, Git, Jenkins, Agile/Scrum

EDUCATION

Master's in Computer Science — Montclair State University

January 2022 – January 2024